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Portfolio Problems for Test 2

Exercise 2.7.1c *(c) Consider the subsequences (s_{2n}) and (s_{2n+1}), and show how the Monotone Convergence Theorem leads to a third proof for the Alternating Series Test.*

Proof. Let (a_n) be a decreasing sequence of positive real numbers with $a_n \to 0$. Consider the two subsequences:

$$s_{2n} = \sum_{k=1}^{2n} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - \cdots - a_{2n}$$

$$s_{2n+1} = \sum_{k=1}^{2n+1} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - \cdots + a_{2n+1}$$

Note that $s_{2(n+1)} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0$, since a_n is decreasing. Hence (s_{2n}) is increasing.

Similarly, $s_{2n+1} - s_{2(n+1)+1} = a_{2n+2} - a_{2n+3} \geq 0$, so (s_{2n+1}) is decreasing.

Also, $s_{2n+1} - s_{2n} = a_{2n+1} \geq 0$, so $s_{2n} \leq s_{2n+1}$. Since $a_n \to 0$, we also have $a_{2n+1} \to 0$, so this difference goes to 0.

Since (s_{2n}) is increasing, and since $s_{2n} \leq s_{2n+1} \leq s_1$, it is bounded above. Similarly, (s_{2n+1}) is decreasing and bounded below by s_2.

By the Monotone Convergence Theorem, both (s_{2n}) and (s_{2n+1}) converge.

Let $\lim s_{2n} = L$ and $\lim s_{2n+1} = M$. Since $s_{2n+1} - s_{2n} = a_{2n+1} \to 0$, it follows that $M - L = 0$, so $L = M$.

Because both the even and odd subsequences converge to the same limit L, the full sequence (s_n) converges to L.

$\therefore \sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.

□

Exercise 3.3.9 *(c) Because $x \in K$, there must exist an open set O_{λ_0} from the original collection that contains x as an element. Explain how this leads to the desired contradiction.*

Proof.

Suppose, for contradiction, that K is not compact.

Since $x \in K$ and $\{O_\lambda\}_{\lambda \in \Lambda}$ covers K, there is some λ_0 such that $x \in O_{\lambda_0}$.

Because O_{λ_0} is open and contains x, there exists $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq O_{\lambda_0}$.

As $|I_n| \to 0$ by part (a), we can choose N so that for all $n \geq N$, $|I_n| < \varepsilon$.

If we write $I_n = [a_n, b_n]$, then since $x \in I_n$ and $b_n - a_n = |I_n| < \varepsilon$, we have

$$a_n > x - \varepsilon \quad \text{and} \quad b_n < x + \varepsilon.$$

Then $I_n \subseteq (x - \varepsilon, x + \varepsilon)$ for all $n \geq N$.

Thus $I_n \subseteq O_{\lambda_0}$ for all $n \geq N$, so $I_n \cap K \subseteq O_{\lambda_0}$. In particular, $I_n \cap K$ is covered by just one open set.

This contradicts part (a), which says $K \cap I_n$ has no finite subcover from $\{O_\lambda\}_{\lambda \in \Lambda}$.

Therefore, our initial assumption that K is not compact leads to a contradiction. Hence every open cover of K has a finite subcover, and K is compact.

$\therefore$ K is compact.

□

Exercise 4.2.3 *(c) Conjecture and verify* $\lim_{x\to 1} t(x)$

Proof.

We conjecture

$$\lim_{x\to 1} t(x) = 0.$$

Let $\varepsilon > 0$. Suppose $t(x) \geq \varepsilon$. Writing $x = \frac{m}{n}$ in lowest terms (with $n \in \mathbb{N}$), we have

$$t(x) = \frac{1}{n} \geq \varepsilon \quad \Rightarrow \quad n \leq \frac{1}{\varepsilon}.$$

Thus, if $t(x) \geq \varepsilon$, the denominator n is bounded above by $\frac{1}{\varepsilon}$. Consider only those x in the interval $(0, 2)$. For each such n, there are only finitely many rational numbers $\frac{m}{n}$ in $(0, 2)$ with this denominator. Hence the set

$$E_\varepsilon = \{x \in (0, 2) : t(x) \geq \varepsilon\}$$

is finite. (Note: $1 \in E_\varepsilon$ because $t(1) = 1 \geq \varepsilon$ for any $\varepsilon \leq 1$; for $\varepsilon > 1$ the set is empty.)

If $\varepsilon > 1$, then E_ε is empty, and the conclusion follows trivially. So we may assume $0 < \varepsilon \leq 1$; in this case E_ε is finite and contains 1.

Since E_ε is finite, let

$$\delta = \min\{1,\ |1 - y| : y \in E_\varepsilon,\ y \neq 1\}.$$

If $E_\varepsilon = \{1\}$, simply take $\delta = 1$. Then $\delta > 0$ and by construction, if $0 < |x - 1| < \delta$, we have $x \notin E_\varepsilon$ and $x \in (0, 2)$.

For any such x:

- If x is irrational, then $t(x) = 0 < \varepsilon$.
- If x is rational, write $x = \frac{m}{n}$ in lowest terms. Then $x \in (0, 2)$ and $x \notin E_\varepsilon$ implies $t(x) < \varepsilon$.

Thus $t(x) < \varepsilon$ whenever $0 < |x - 1| < \delta$.

$\therefore \lim_{x\to 1} t(x) = 0.$

□

Exercise 4.5.3. *A function f is increasing on A if $f(x) \leq f(y)$ for all $x < y$ in A. Show that if f is increasing on $[a, b]$ and satisfies the Intermediate Value Property (Definition 4.5.3), then f is continuous on $[a, b]$.*

Proof.

Let $c \in [a, b]$. We show f is continuous at c.

Case 1: $f(a) < f(c) < f(b)$**.**

Let $\varepsilon > 0$. Without loss of generality, assume

$$\varepsilon < \min\{f(c) - f(a), f(b) - f(c)\}.$$

Since any $\varepsilon > 0$ can be reduced to satisfy this bound, proving continuity for such ε is sufficient. Then,

$$f(c) - \frac{\varepsilon}{2} > f(a) \quad \text{and} \quad f(c) + \frac{\varepsilon}{2} < f(b).$$

Since f satisfies the Intermediate Value Property, there exist

$$c_1 \in (a, c) \quad \text{with} \quad f(c_1) = f(c) - \frac{\varepsilon}{2}$$

and

$$c_2 \in (c, b) \quad \text{with} \quad f(c_2) = f(c) + \frac{\varepsilon}{2}.$$

Because f is increasing, for any x with $c_1 < x < c_2$ we have

$$f(c_1) \leq f(x) \leq f(c_2).$$

Thus

$$f(c) - \frac{\varepsilon}{2} \leq f(x) \leq f(c) + \frac{\varepsilon}{2},$$

so $|f(x) - f(c)| \leq \frac{\varepsilon}{2} < \varepsilon$.

Now set $\delta = \min\{c - c_1, c_2 - c\}$. If $|x - c| < \delta$, then $c_1 < x < c_2$, and therefore $|f(x) - f(c)| < \varepsilon$. Hence f is continuous at c.

Case 2: $f(c) = f(a)$**.**

Let $\varepsilon > 0$.

If $f(b) = f(c)$, monotonicity implies f is constant on $[c, b]$, so $|f(x) - f(c)| = 0$ for $x \geq c$. For $x \leq c$, we will show below that $f(x) = f(c)$. Thus f is continuous at c in this subcase. So assume $f(b) > f(c)$.

Let $\varepsilon' = \min\{\varepsilon,\ 2(f(b) - f(c))\}$. Then $\varepsilon' > 0$ and $f(c) + \varepsilon'/2 \leq f(b)$. By the Intermediate Value Property on $[c, b]$, there exists $c_2 \in (c, b)$ with

$$f(c_2) = f(c) + \frac{\varepsilon'}{2}.$$

For any $x \in (c, c_2)$, monotonicity gives $f(c) \leq f(x) \leq f(c_2)$, so

$$|f(x) - f(c)| \leq \frac{\varepsilon'}{2} < \varepsilon'.$$

For $x \in [a, c)$, monotonicity gives $f(x) \leq f(c)$. Since $f(c) = f(a)$, we actually have

$$f(a) \leq f(x) \leq f(c) = f(a),$$

which makes $f(x) = f(a) = f(c)$. Hence f is constant on $[a, c]$, and $|f(x) - f(c)| = 0 < \varepsilon$ for all $x \in [a, c]$.

Set $\delta = c_2 - c$. If $|x - c| < \delta$ and $x \geq c$, then $x \in (c, c_2)$ and $|f(x) - f(c)| < \varepsilon$. If $x \leq c$, then $f(x) = f(c)$, so $|f(x) - f(c)| = 0 < \varepsilon$. Thus f is continuous at c.

Case 3: $f(c) = f(b)$**.**

Let $\varepsilon > 0$.

If $f(a) = f(c)$, then f is constant on $[a, c]$ by monotonicity, so $|f(x) - f(c)| = 0$ for $x \leq c$. For $x \geq c$, we will show $f(x) = f(c)$. Hence continuity holds in this subcase. So assume $f(a) < f(c)$.

The argument is symmetric to Case 2. Let $\varepsilon' = \min\{\varepsilon,\ 2(f(c) - f(a))\}$. Then $\varepsilon' > 0$ and $f(c) - \varepsilon'/2 \geq f(a)$. By the Intermediate Value Property on $[a, c]$, there exists $c_1 \in (a, c)$ with

$$f(c_1) = f(c) - \frac{\varepsilon'}{2}.$$

For any $x \in (c_1, c)$, monotonicity gives $f(c_1) \leq f(x) \leq f(c)$, so

$$|f(x) - f(c)| \leq \frac{\varepsilon'}{2} < \varepsilon'.$$

For any $x \in (c, b]$, since f is increasing, $f(c) \leq f(x) \leq f(b) = f(c)$, so $f(x) = f(c)$. Thus $|f(x) - f(c)| = 0 < \varepsilon$ for all $x \in [c, b]$.

Take $\delta = c - c_1$. If $|x - c| < \delta$ and $x \leq c$, then $x \in (c_1, c)$ and $|f(x) - f(c)| < \varepsilon$. If $x \geq c$, then $x \in [c, b]$ and $f(x) = f(c)$, so $|f(x) - f(c)| = 0 < \varepsilon$. Hence f is continuous at c.

$\therefore$ Since $c \in [a, b]$ was arbitrary, f is continuous on $[a, b]$.

□